

Addis Ababa University
Department of Physics
General Physics Mid Exam (30%)

Name : _____

Time : 2 hrs

I.D. No. _____ **Section :** _____

Part I : Choose the correct answer from the given alternatives and put your answers on the space provided at the end of the questions (each 1.5 pts)

1. You're driving your car towards an intersection. A Porsche is stopped at the red light. You're travelling at 40 km/h (11.11 m/s). As you are 15 m from the light, the light turns green, and the Porsche accelerates from rest at 3 m/s^2 . You continue at constant speed. The Porsche will be pulled over. How far from the stop line do you pass the Porsche? At what time, measured from when the light turned green, do you pass the Porsche?

- (A) 5.63 s and 0.106 km (B) 1.78 s and 4.75 m
 (C) 5.63 s and 47.5 m (D) 26.4 s and 1.045 km (E) 0.266 s and 0.106 m

2. (from question number 1) : As the Porsche keeps accelerating, it eventually catches up to you again. How far from the stop line does it pass you? At what time, measured from when the light turned green, does it pass you?

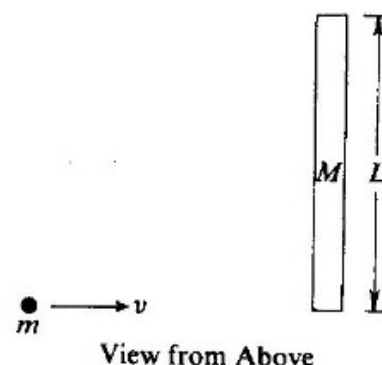
- (A) 5.63 s and 0.106 km (B) 1.78 s and 4.75 m
 (C) 5.63 s and 47.5 m (D) 26.4 s and 1.045 km (E) 0.266 s and 0.106 m

3. The potential energy of a body constrained to move on a straight line is kx^4 where k is a constant. The position of the body is x , its speed v , and its mass b . The force of the body is.

- (A) $(1/2)mv^2$ (B) $-4kx^3$ (C) kx^4 (D) $-(kx^5)/5$ (E) mg

4. A uniform stick of length L and mass M lies on a frictionless horizontal surface. A point particle of mass m approaches the stick with speed v on a straight line perpendicular to the stick that intersects the stick at one end, as shown above. After the collision, which is elastic, the particle is at rest. The speed V of the center of mass of the stick after the collision is

- (A) $(m/M)v$ (B) $[m/(M + m)]v$ (C) $[(m/M)^{1/2}]v$
 (D) $[[m/(M + m)]^{1/2}]v$ (E) $3m/Mv$

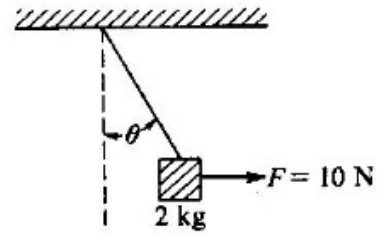


5. An exceptional standing jump would raise a person 0.80 m off the ground. To do this, a 61 kg person crouches 0.20 m and pushes off from the ground, exerting a force on it. By Newton's third law, the ground pushes back with an equal and opposite force, which accelerates the person off the ground. What is the jumper's speed just as he leaves the ground?

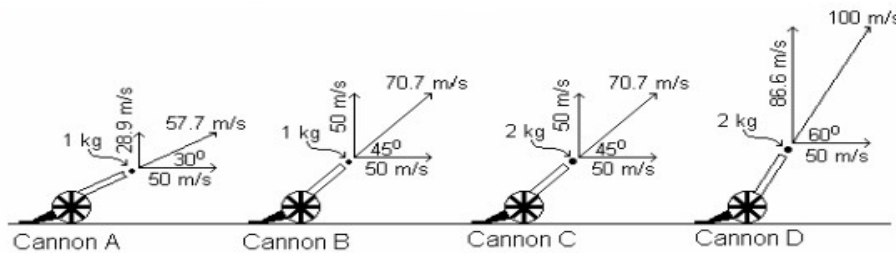
- (A) $2\sqrt{8} \text{ m/s}$ (B) 16 m/s (C) 12 m/s (D) 4 m/s (E) 1.6 m/s

6. A 2-kilogram box hangs by a massless rope from a ceiling as shown. A force slowly pulls the box horizontally to the side until the force is equal to 10 N. The box is then in equilibrium. The angle that the rope makes with the vertical is closest to:

- (A) $\sin^{-1}(0.5)$ (B) $\tan^{-1}(2.0)$ (C) $\sin^{-1}(2.0)$
 (D) 45° (E) $\tan^{-1}(0.5)$



7. The diagram below shows four cannons firing shells with different masses at different angles of elevation. The horizontal component of the shell's velocity is the same in all four cases. In which case will the shell have the greatest range if air resistance is neglected?

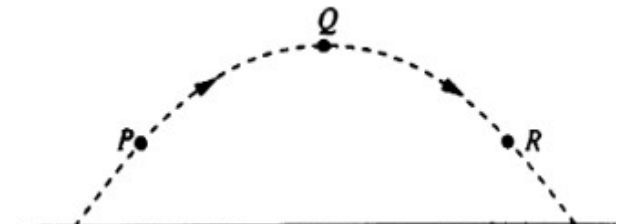


- (A) cannon A
 (B) cannon B only
 (C) cannon C only
 (D) cannon D

8. A diver initially moving horizontally with speed v dives off the edge of a vertical cliff and lands in the water a distance d from the base of the cliff. How far from the base of the cliff would the diver have landed if the diver initially had been moving horizontally with speed $2v$?

- (A) d (B) $(2d)^{1/2}$ (C) $2d$ (D) $4d$

9. A ball is thrown and follows the parabolic path shown above. Air friction is negligible. Point Q is the highest point on the path. Points P and R are the same height above the ground. How do the speeds of the ball at the three points compare?



- (A) $v_P < v_Q < v_R$ (B) $v_R < v_Q < v_P$ (C) $v_Q < v_R < v_P$ (D) $v_Q < v_P = v_R$

10. The driver of a train travelling at 40 m/s applies the brakes as the train enters a station the train slows down at a rate of 2 m/s^2 . The plat form is 400 m long. will the train stop in time?

- (a) No, it needs 2 m longer plat form (b) No, it needs 2.5 m longer plat form
 (c) Yes, the train will stop in time (d) Yes, it will stop 1.75 m before it reaches end of the plat form

Answer box for Part I

1		2		3		4		5	
6		7		8		9		10	

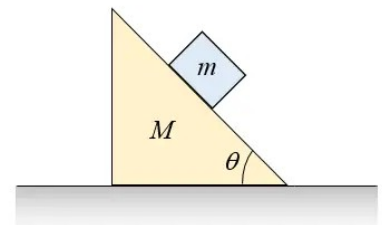
Part II : Work Out (each 3 pts)

1. In a series of successive measurements in an experiment, the readings of the period of oscillation of a simple pendulum were found to be 2.63s, 2.56 s, 2.42s, 2.71s and 2.80s. Calculate

- (i) the mean value of the period of oscillation
- (ii) the absolute error in each measurement
- (iii) the mean absolute error (rounded of 2nd decimal place)
- (iv) the relative error
- (v) the percentage error. Express the result in proper form.

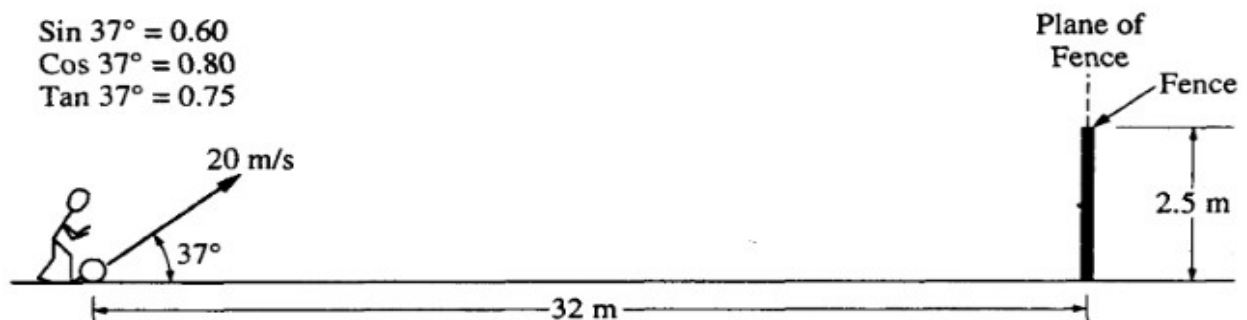
2. An object is traveling in a straight line. Its acceleration is given by : $a = Ct^n$ where C is a constant, n is a real number, and t is time. Find the general equations for the position and velocity of the object as a function of time.

3. A block of mass m is placed on an incline of mass M , as shown in the figure below. The coefficient of kinetic friction between the block and incline is μ_k . What is the minimum coefficient of static friction between the incline and floor, so that the incline doesn't slide on the floor when the block slides down the incline?



4. A ball of mass 0.1 kg bounces on a hard surface. Every time it hits the floor, it loses a quarter of its kinetic energy. If the ball is released from a height of 1.00 m, after how many bounces will the ball bounce no higher than 0.25 m?

5. A ball of mass 0.5 kilogram, initially at rest, is kicked directly toward a fence from a point 32 meters away, as shown below. The velocity of the ball as it leaves the kicker's foot is 20 meters per second at an angle of 37° above the horizontal. The top of the fence is 2.5 meters high. The ball hits nothing while in flight and air resistance is negligible.



Note: Diagram not drawn to scale.

- a. Determine the time it takes for the ball to reach the plane of the fence.
- b. Will the ball hit the fence? If so, how far below the top of the fence will it hit? If not, how far above the top of the fence will it pass?
- c. On the axes below, sketch the horizontal and vertical components of the velocity of the ball as functions of time until the ball reaches the plane of the fence.

